

Sample Size for a Two-Sample t-Test

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Introduction

Before running a two-group comparison, the investigator needs to know how many observations per group will be required to detect a difference of scientific importance with acceptable probability. Power analysis for the two-sample t-test answers this question by linking four quantities: the true effect size, the significance level, the power, and the sample size. Given any three, R can solve for the fourth.

Prerequisites

Familiarity with the two-sample t-test and with the concepts of type I and type II error is assumed. The reader should know that “power” means $1 - \beta$, the probability of rejecting H_0 when H_1 is true.

Theory

Under H_0 , the t statistic follows a central t distribution with $\nu = n_1 + n_2 - 2$ degrees of freedom (Student’s test, equal variances). Under H_1 where $\mu_1 - \mu_2 = \delta$, the statistic follows a *non-central* t distribution with non-centrality parameter

$$\lambda = \frac{\delta}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}.$$

Power is the probability that the test statistic exceeds the critical value under this non-central distribution. For Cohen’s standardised effect size $d = \delta/\sigma$ and equal group sizes $n_1 = n_2 = n$, this simplifies to $\lambda = d\sqrt{n/2}$.

The four power quantities and their trade-offs:

- **Effect size d :** the minimum standardised difference considered scientifically meaningful. By Cohen’s conventions, $d = 0.2$ is small, 0.5 is medium, 0.8 is large – but these conventions are field-dependent and should be used with judgement.
- **Significance level α :** usually 0.05 for two-sided testing.
- **Power:** conventionally 0.80 for exploratory work, 0.90 for confirmatory or registered trials.
- **Sample size per group n :** what we typically solve for.

Assumptions

Power analysis inherits the assumptions of the t-test it plans for: independent groups, approximately normal data (or large enough n for the CLT), and equal variances for Student’s test (unequal for Welch’s).

R Implementation

```
library(pwr)

pwr.t.test(d = 0.5, sig.level = 0.05, power = 0.80,
           type = "two.sample", alternative = "two.sided")
```

```
pwr.t.test(n = 30, sig.level = 0.05, power = 0.80,
           type = "two.sample", alternative = "two.sided")

pwr.t.test(n = 30, d = 0.5, sig.level = 0.05,
           type = "two.sample", alternative = "two.sided")

library(pwrss)
pwrss.t.2means(mu1 = 110, mu2 = 100, sd1 = 18, sd2 = 15,
               power = 0.80, alpha = 0.05)
```

The first `pwr.t.test()` call asks: how many subjects per group do I need to detect a medium-sized effect ($d = 0.5$) with 80% power? The second asks: given $n = 30$ per group, what is the minimum detectable effect at 80% power? The third asks: given $n = 30$ and $d = 0.5$, what is my power? The `pwrss::pwrss.t.2means()` call works directly in raw-mean terms, which is often more natural when planning a clinical trial.

Output & Results

The first call returns $n \approx 64$ per group – 128 subjects in total. The second returns a minimum detectable $d \approx 0.735$, corresponding to a raw difference of about 11 points if $\sigma = 15$. The third returns a power of approximately 0.48, showing that 30 per group is badly underpowered for a medium effect.

Two-sample t test power calculation

```
      n = 63.76576
      d = 0.5
sig.level = 0.05
  power = 0.8
alternative = two.sided
```

Interpretation

The recommended manuscript phrasing is: “Assuming a between-group standardised mean difference of $d = 0.5$ (a medium effect by Cohen’s criteria), a two-sided α of 0.05, and power of 0.80, the required sample size is 64 per group. Allowing for 10% loss to follow-up, we aim to enrol 72 per group (144 total).” The power analysis should be reported in the methods section of the grant or protocol, not buried in an appendix.

Practical Tips

- Base your effect size on published data or a pilot, not on a convention like $d = 0.5$. If the published effect is $d = 0.3$, planning for $d = 0.5$ will leave you underpowered.
- Use Welch’s sample size for unequal variances (`pwr.t2n.test()` or `pwrss.t.2means()` with `sd1 != sd2`). It is slightly more conservative, which is what you want at the planning stage.
- Plan for loss to follow-up, noncompliance, and data exclusion by inflating the computed n by a realistic percentage (typically 10-25%).
- If the minimum clinically important difference (MCID) is known in raw units, use that directly rather than translating to d . It is more interpretable to clinicians.
- Always produce a sensitivity analysis: a curve of power against n , or a tornado plot of sample size against plausible values of effect size and SD.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.

- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Sample size, power, and Quarto reporting
- Power: closed-form and simulation